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Govt. of Telangana

Agricultural Extension Advisory Bulletin | Regional Agricultural Research Station

Bulletin ID: PJTSAU-EXT-AGRI-2024-V1 • **Season Cycle:** Kharif / Rabi Production Guidelines •

Districts: Warangal, Karimnagar, Khammam, Nalgonda, Adilabad

1. AGRO-CLIMATIC ADVISORY OVERVIEW

This advisory bulletin provides actionable, research-backed management practices for commercial smallholder farmers across Telangana. Due to changing rainfall patterns and temperature fluctuations, farmers must prioritize balanced nutrition, timely threshold-based pest control, and active moisture conservation to safeguard crop productivity and farm economics.

2. COTTON (KAPAS) CROP MANAGEMENT PACKAGE

Cotton is a primary commercial cash crop cultivated across the black cotton and red loamy soils of Telangana. Sucking pests during the vegetative phase and bollworm incidence during flowering and boll formation cause substantial economic damage if not intervened at the Economic Threshold Level (ETL).

A. Protocol — Pink Bollworm *(Pectinophora gossypiella)*

- **ETL:** 8 rosette flowers per 100 plants or 10% damaged green bolls across the field.
- **Monitoring & Trapping:** Install yellow/pheromone traps with Gossyplure septa at 8 traps/acre at 45 days after sowing (DAS) for continuous field surveillance.
- **Chemical Intervention:** When crossing ETL, spray **Profenophos 50% EC @ 2 ml/L** of water or **Emamectin Benzoate 5% SG @ 0.5 g/L** of water.
- **Safety Warning:** Avoid repeated sprays of synthetic pyrethroids; they induce secondary resurgence of aphids and whiteflies.

B. Protocol — Sucking Pest Complex (Aphids, Jassids, Thrips)

- **Symptoms:** Downward or upward leaf curling, yellow margins (hopper burn), and stunted terminal growth during dry spells.
- **Chemical Management:** Spray **Flonicamid 50% WG @ 0.3 g/L** of water or **Dinotefuran 20% SG @ 0.4 g/L** of water.
- **Eco-Friendly & Biological Measure:** For preventive and early-stage control, spray **5% Neem Seed Kernel Extract (NSKE)** or cold-pressed Neem Oil (1500 ppm) @ 5 ml/L of water.

C. Nitrogen and Potash Split Fertilizer Schedule

Crop Stage / Timing	Fertilizer Grade & Dosage (per Acre)	Placement & Operational Instructions
Basal Dose At Sowing	100 kg Single Super Phosphate (SSP) + 15 kg Muriate of Potash (MOP)	Band placement 5 cm below seed depth to optimize phosphorus fixation.
First Top Dressing 20 to 25 DAS	25 kg Urea + 10 kg Muriate of Potash (MOP)	Apply under adequate root-zone soil moisture conditions.
Second Top Dressing 40 to 45 DAS	30 kg Urea	Side-dress followed immediately by light inter-cultivation or hoeing.
Third Top Dressing 60 to 65 DAS	25 kg Urea + 10 kg Muriate of Potash (MOP)	Broadcast along crop rows just prior to peak flowering phase.

3. CHILLI (MIRAPA) INTEGRATED PROTECTION PACKAGE

Chilli cultivation across Khammam, Mahabubabad, and Warangal requires intensive pest and fungal surveillance to curb yield drops triggered by invasive vector attacks and fruit rots.

A. Vector **Invasive Black Thrips *(Thrips parvispinus)***

- **Damage:** Nymphs and adults feed on floral organs and tender leaves, causing blossom drop, flower drying, and upward curling.
- **Physical Trapping:** Mount blue sticky traps @ 25 to 30 traps/acre at canopy height during seedling establishment.
- **Targeted Spray Schedule:** When nymph population exceeds 5 thrips/flower, spray **Spinetoram 11.7% SC @ 1 ml/L**, **Cyantraniliprole 10.26% OD @ 1.2 ml/L**, or **Fipronil 5% SC @ 2 ml/L**.
- **Resistance Strategy:** Rotate chemical groups with distinct modes of action across sprays; never apply identical chemical class back-to-back.

B. Fungal **Anthracnose, Die-Back & Fruit Rot**

- **Pathogen:** *Colletotrichum capsici*.
- **Symptoms:** Dark, circular sunken lesions with concentric black rings on ripening fruit pods and drying back of twigs downwards.
- **Fungicidal Spray:** Spray **Azoxystrobin 18.2% + Difenoconazole 11.4% SC @ 1 ml/L** or **Copper Oxychloride 50% WP @ 3 g/L** upon initial symptom appearance.
- **Application Technique:** Ensure high-pressure, uniform drenching covering upper and lower foliage surfaces.

4. GOVERNMENT WELFARE AND IRRIGATION SUBSIDY SCHEMES

Rythu Bharosa (Farmer Investment Support)

Direct Benefit Transfer

Objective: Direct financial support to help farmers procure seeds, fertilizers, and field inputs without non-institutional debt.
Target Beneficiaries: Eligible small, marginal, and tenant farmers holding verified agricultural titles.
Documentation: Aadhaar Card, updated Pattadar Passbook recorded on Dharani portal, and active Aadhaar-linked bank account.

Telangana State Micro Irrigation Project (TSMIP)

Water Conservation

Objective: Promotes water conservation and yield enhancement through subsidized drip and micro-sprinkler systems.
Subsidy Tiers: 100% subsidy for SC/ST farmers; 90% for small & marginal farmers (≤ 5 acres); 80% for other eligible categories.
Documentation: Land Title Deed, Soil & Irrigation Water Quality Test Certificate, Certificate from Tahsildar / Mandal Revenue Office.

Rythu Bima (Farmers Life Insurance Scheme)

Social Security

Objective: Comprehensive financial security to agrarian households in the event of the breadwinner's casualty.
Coverage & Eligibility: Fixed lump-sum compensation of **₹5,00,000 INR** deposited directly into nominee's account within 10 days of claim approval. Enrolled landholders aged 18 to 59 years.
Documentation: Pattadar Passbook, Nominee Aadhaar Card, Age Proof, and Village Revenue Officer (VRO) verification.

5. AGRONOMIC ADVISORY NOTICE & COMPLIANCE

- **Dosage Precision:** Chemical sprays must follow university-approved concentrations strictly. Under-dosing promotes rapid insect resistance, while over-dosing causes phytotoxicity and leaf scorching.
- **Weather Precautions:** Avoid applying chemical sprays during strong wind conditions or immediately preceding heavy rainfall forecasts.
- **Field Extension Support:** For unresolved field abnormalities or localized disease symptoms, farmers are advised to consult their local Mandal Agricultural Officer (MAO) or visit the nearest Krishi Vigyan Kendra (KVK).